

ENONTEKIO KILPISJARVI, Finland

WMO: 028010

Lat: 69.049N

Lon: 20.791E

Elev: 1584

StdP: 13.87

Time Zone: 2.00 (EUE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.3	-21.7	-32.2	0.9	-26.1	-27.1	1.3	-21.6	31.5	23.6	28.4	21.7	1.3	150	0.829

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	69.8	58.6	65.7	56.4	62.0	54.4	59.8	67.4	57.6	63.8	55.6	60.8	8.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.5	72.2	62.2	54.5	66.9	60.6	52.6	62.5	58.6	26.9	67.6	25.5	63.8	24.2	61.0	68.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.6	22.7	19.6	DB	-32.0	74.4	3.8	3.8	-34.8	77.1	-37.0	79.4	-39.1	81.5	-41.9	84.3	(4)
(5)			WB	-32.1	62.9	3.7	2.0	-34.7	64.4	-36.8	65.5	-38.9	66.7	-41.5	68.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.7	8.8	9.4	14.9	25.5	37.0	45.7	53.5	50.7	43.0	31.8	20.5	13.5		
	DBStd	18.46	14.45	14.38	11.91	8.86	6.33	5.86	5.72	5.47	5.82	7.60	10.32	13.91		
	HDD50	7680	1276	1137	1089	734	405	154	29	57	218	565	885	1131		
	HDD65	12901	1741	1557	1554	1184	867	578	358	442	659	1030	1335	1596		
	CDD50	256	0	0	0	0	3	26	137	81	9	0	0	0		
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0		
	CDH74	22	0	0	0	0	1	1	18	2	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	8.3	7.9	7.9	7.9	8.1	8.5	8.7	8.6	8.4	8.9	8.4	8.2	8.4		
	PrecAvg	18.8	2.0	1.4	1.2	0.8	1.0	1.6	2.7	1.9	1.3	1.7	1.3	1.6		
(15) Precipitation	PrecMax	24.2	7.2	2.8	5.2	1.9	2.9	3.7	7.0	3.3	3.1	4.5	3.5	4.1		
	PrecMin	11.0	0.5	0.2	0.3	0.1	0.2	0.1	0.4	0.9	0.3	0.6	0.2	0.0		
	PrecStd	3.9	1.5	0.7	1.2	0.5	0.6	1.0	1.5	0.6	0.8	0.9	0.8	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	38.8	40.6	46.9	66.0	69.1	76.2	71.6	60.5	50.1	42.3	40.4		
		MCWB	35.7	35.7	35.4	41.0	51.4	55.4	61.4	59.6	52.6	45.8	39.0	37.1		
	2%	DB	34.9	35.5	36.3	42.6	56.5	63.4	71.9	67.5	56.9	47.4	39.2	36.8		
		MCWB	32.7	32.5	33.4	37.5	46.4	52.7	59.6	58.6	51.1	44.4	37.1	34.3		
	5%	DB	31.4	32.5	33.5	40.1	50.2	59.8	67.8	63.1	54.1	45.0	35.4	33.1		
		MCWB	29.6	30.6	31.2	35.8	43.1	50.6	58.3	56.1	49.4	42.1	33.7	31.0		
	10%	DB	27.9	28.8	30.9	37.8	46.4	56.5	63.7	59.7	51.6	42.4	32.8	30.6		
		MCWB	26.5	27.3	28.7	34.3	40.8	49.3	55.7	54.2	47.7	40.0	31.4	28.9		
	0.4%	WB	35.9	36.4	35.9	41.1	52.2	57.5	63.7	62.4	54.4	46.7	39.7	37.6		
		MCDB	38.2	38.8	39.4	46.2	65.3	64.3	72.8	69.0	58.5	49.2	41.7	39.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.0	32.8	33.6	38.0	46.6	54.6	60.8	59.1	51.9	44.5	37.1	34.5		
		MCDB	34.7	35.0	36.1	41.7	54.7	61.0	69.9	65.2	56.0	47.0	39.2	36.7		
	5%	WB	29.6	30.5	31.4	36.2	43.6	52.1	58.7	56.8	50.1	42.4	33.9	31.2		
		MCDB	31.5	32.5	33.4	39.6	49.6	58.6	65.8	62.3	53.5	44.9	35.2	32.9		
	10%	WB	26.6	27.3	28.8	34.3	41.2	49.8	56.8	54.7	48.2	40.0	31.4	28.9		
		MCDB	28.0	28.7	30.9	37.6	45.7	55.6	62.7	59.3	51.3	42.4	32.7	30.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	15.0	16.9	16.0	12.1	12.9	13.6	12.4	10.1	8.0	10.8	14.1		
		MCDBR	13.9	13.9	13.1	14.3	19.6	20.4	21.1	18.4	14.3	9.4	10.3	11.7		
		MCWBR	12.9	12.9	11.3	10.5	11.8	11.4	11.1	10.6	9.5	7.5	9.0	10.7		
	5% WB	MCDBR	13.8	13.5	13.0	12.6	19.0	17.9	18.2	16.8	12.7	9.1	10.0	12.1		
		MCWBR	12.9	12.6	11.4	9.5	11.6	10.9	10.2	10.3	8.9	7.5	8.9	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.183	0.235	0.281	0.304	0.321	0.325	0.330	0.317	0.288	0.245	0.183	N/A		
	taud		1.902	2.061	2.139	2.189	2.291	2.468	2.491	2.509	2.547	2.216	1.952	N/A		
	Ebn at Noon		15	167	233	264	274	277	272	263	243	164	12	0		
	Edh at Noon		2	14	25	32	33	29	28	24	17	11	1	0		
(44) All-Sky Solar Radiation	RadAvg		6	123	562	1188	1587	1565	1446	1026	547	169	21	0		
	RadStd		1	14	33	63	128	126	127	78	50	20	2	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page